

## Query Review 1

Original Query:

```
1 | SELECT e.first_name, e.last_name, d.department_name
2 | FROM employees e
3 | JOIN departments d
4 | ON e.employee_id = d.department_id;
```

Correct, but the JOIN condition is wrong.

e.employee\_id = d.department\_id --> There is no relationship between them.

Correct Query:

```
1 | SELECT e.first_name, e.last_name, d.department_name
2 | FROM employees e
3 | INNER JOIN departments d
4 | ON e.department_id = d.department_id;
```

## Query Review 2

Original Query:

```
1 | SELECT e.first_name, e.last_name, j.job_title
2 | FROM employees e
3 | JOIN jobs j
4 | ON e.employee_id = j.job_id;
```

JOIN condition is incorrect.

e.employee\_id = j.job\_id --> These columns have different meanings and data types.

Correct Query:

```
1 | SELECT e.first_name, e.last_name, j.job_title
2 | FROM employees e
3 | INNER JOIN jobs j
4 | ON e.job_id = j.job_id;
```

## Query Review 3

Original Query:

```
1 SELECT e.first_name, d.department_name
2 FROM employees e
3 LEFT JOIN departments d
4 ON e.department_id = d.department_id
5 WHERE d.department_name = 'Sales';
```

Logical problem.

LEFT JOIN --> Return all employees, even if they do not belong to a department.

But WHERE d.department\_name = 'Sales' --> filters out all NULL department rows.

So converts the LEFT JOIN into an INNER JOIN.

Correct Query:

Improved Version (if the goal is Sales employees):

```
1 SELECT e.first_name, d.department_name
2 FROM employees e
3 INNER JOIN departments d
4 ON e.department_id = d.department_id
5 WHERE d.department_name = 'Sales';
```

## Query Review 4

Original Query:

```
1 SELECT e.first_name, m.first_name AS manager_name
2 FROM employees e
3 JOIN employees m
4 ON e.employee_id = m.manager_id;
```

This is a self-join and the syntax is valid.

employee.employee\_id = manager.manager\_id --> Find managers whose manager\_id points to an employee.

Correct Query:

```
1 SELECT e.first_name, m.first_name AS manager_name
2 FROM employees e
3 INNER JOIN employees m
4 ON e.manager_id = m.employee_id;
```